

# The Normative Pull of Prediction Markets on Our Beliefs and Credences

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## 1 Introduction

The market price of a bet on “Zohran Mamdani will win the New York City mayoral election” on Polymarket (“New York City Mayoral Election” 2025) is currently (July 7 2025) 69¢ — nice.

What should this mean for our doxastic and credal attitudes towards the proposition that Mamdani will win? Is the market price being what it is a reason to believe it? Would it be a reason to believe it if it were higher? Is it a reason to believe a proposition like “The chance of Mamdani winning is 69%”? To set your credence to 69% (or to any other specific or unspecific value)?

These are the questions I will be addressing in this paper; or, more precisely, I will be discussing the general question(s): In the context of a prediction market, let a bet on a proposition  $q$  be an asset  $b_q$  that pays \$1 if  $q$  and \$0 if  $\neg q$ ; and let  $p_q$  be the market price of such a bet in dollars. What—if anything—is wrong with me if my credence in  $q$  is different from  $p_q$ ? And is there a threshold such that there is something wrong with me if I fail to believe  $q$  despite the price of a bet on  $q$  being above that threshold?

I will sketch a way of thinking about beliefs and credences—inspired by and building upon MacFarlane (2025)—and use it to argue for the following answers:

Sometimes, it can be wrong for one’s credence in  $q$  to deviate from the market price of  $b_q$ . One such case is an agent who believes in one form or another of the efficient market hypothesis (and believes that the available prediction markets for  $q$  are such that it applies), and also believes that they do not possess exclusive information of the relevant kind (where the relevant kind depends on the specific form of the EMH the agent believes).

Clearly, if the threshold view or the Lockean thesis (Jackson 2020, 2–4) about the (metaphysical and normative, respectively) relationship between belief and credence is true, this will directly entail that market prices of bets likewise exert a normative pull on an agent’s beliefs. But the view I will be using does not have this feature. On the contrary, I will use it to argue that it is almost never the case that the market price of a bet on  $q$  is on its own a reason in favour of believing (or disbelieving)  $q$  (or  $\neg q$ ).

Before I give a brief summary of the arguments I will be making, a brief note on terminology: A lot of the argument of this paper turns on the distinction between two normativities, the normativity governing moves in the language game of giving and asking for reasons, and the normativity governing actions properly guided by decision theory (which, I will assume, are exactly those actions for which both doing them, and refraining from doing them, is permitted). I will call the first by the name of “reasonableness”, and the second by the name of “rationality\*<sup>1</sup>”. I hesitate to call it “rationality”, because rationality may have more expansive requirements, or may apply also to other realms of human activity. Hence, it will be “rationality\*” in this paper<sup>1</sup>.

My first argument is in support of the positive conclusion that credences sometimes need to track market prices of bets:

### 1.0.1 Argument 1

1. An agent’s credences are irrational\* if they decision-theoretically recommend irrational\* actions.
2. Sometimes, betting against the market is irrational\*.
3.  $\therefore$  Sometimes, an agent’s credences need to be such as to not recommend betting against the market.
4. Credences which differ from the market price of a bet on  $q$  recommend a bet against the market.
5.  $\therefore$  Sometimes, an agent’s credence in  $q$  needs to be identical to the market price of a bet on  $q$ .

For the negative conclusion that it is almost never the case that the market price of a bet on  $q$  is on its own a reason to believe  $q$ , I give the following argument. Let  $t$  be some threshold for which we might think that  $p_q > t$  gives an agent a decisive reason to believe  $q$ .

### 1.0.2 Argument 2

1. If  $p_q > t$  is on its own a reason to believe  $q$ , then an agent who accepts  $p_q > t$  should by default be willing to also accept  $q$  as a reason for further claims or attitudes.
2. If  $p_q > t$  is the only evidence available to an agent, it will in many cases be wrong for her to accept  $q$  as a reason for further claims or attitudes.
3. Therefore,  $p_q > t$  not on its own a reason to believe  $q$ .

The core of these arguments is based on the aforementioned view on the relationship between belief and credence based on MacFarlane (2025). It will be developed in section Section 2, which will also make clear why and how it motivates premises 1 and 4 in the first argument and premise 1 in the second. The remaining premise (premise 2) of argument 1 will be discussed in section Section 4.1. Premise 2 of

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<sup>1</sup>As MacFarlane, who I am following here, writes—though he sticks with using the word „rationality“ even in the face of such considerations: ” I am rejecting a long philosophical tradition of thinking of rationality as the faculty for apprehending and responding to reasons. I’m even going against etymology, which derives ‘rational’ from ratio, reason. But one kind of philosophical progress is the recognition that there are two distinct things which we confused before” (MacFarlane 2025, 857).

argument 2 will be argued for in section Section 3 .

## 2 Belief and Credence

I will be thinking about the concepts of “belief” and “credence” in a methodologically behaviourist way here, that is, as theoretical terms of psychology which are to be introduced in terms of vocabulary pertaining to overt behaviour<sup>2</sup>. As such, they need to earn their keep by helping us make sense of, explain and predict the behaviour of human beings. The question to ask, then (along the lines of MacFarlane (2025) who this analysis is based on), is: What talents do these concepts bring to the table, and what job should they be given in our psychological theories?

I will start in section Section 2.1 by showcasing the utility of the concept of “credence” in describing and explaining human behaviour. I will then move on to describe the two distinctive talents which the concept of “belief” has above and beyond what “credence” can do, and introducing the specific job which MacFarlane has in mind for it in within our theorising of the human being and it’s actions in section Section 2.2. The result will be a form of belief-credence dualism, whose explanatory power, as well as perhaps counter-intuitive consequences will be showcased in applying it to the “Sleeping Beauty” problem in section Section 2.4.

Section Section 2.5 will raise the question of whether perhaps this dualism has pulled belief and credence so far apart as to make unintelligible how they interact. It will answer this question by turning to the practice of betting as a bridge between the two attitudes.

### 2.1 The Power of Credence

Why do we even need credence in our theoretical vocabulary? Isn’t belief enough?

The following example from MacFarlane (2025) shows the limitations of belief in explaining human action:

Sam is walking on a narrow board that spans a shallow ditch. Why is he doing that? Because his lunch is on the other side of the ditch, and he believes that by walking across the board, he will get there. His action is rational in light of his goals and beliefs, and thus we explain it. But now let’s consider a similar case, but let the board cross a chasm one hundred meters deep. Now Sam does not cross. Why not? He still wants to get to the other side, and he still believes that by walking across the board, he will get there. Yet, if we make the gap deep enough, it ceases to be rational for him to cross. (MacFarlane 2025, 848)

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<sup>2</sup>In this (and some other) ways, the analysis here bears resemblance to the one Wilfrid Sellars gives of “thoughts” as inner episodes in §§46-59 of “Empiricism and the Philosophy of Mind” (Sellars 1997).

In simple cases, it looks like beliefs and desires do explain (individual) behaviour; but as the example shows, the (binary) concept of belief is lacking in the quantitative structure needed to account for some of the relevant differences in the modified situation. Credence-based decision theory, on the other hand, is able to capture these excellently:

“Suppose Sam’s utilities for the various possible outcomes are as follows:

| Outcome                             | Utility |
|-------------------------------------|---------|
| Crosses successfully and gets lunch | 1       |
| Stays put and skips lunch           | -1      |
| Falls into shallow ditch            | -1      |
| Falls into deep chasm               | -1000   |

Then, if Sam’s credence that he will get across the board without falling is 0.99, the expected utility of crossing is 0.98 in ‘Shallow’ and -9.01 in ‘Deep’, while the expected utility of not crossing is -1 in both cases.” (MacFarlane 2025, 848)

Consequently, what the role of credence in or psychological theorising should be, is the following: We describe an agent as having a certain set of credences iff those credences in fact decision-theoretically rationalise and explain their actions: Credences are those internal states which disposes one to make the decision which maximises the utility one expects based them.

## 2.2 Belief’s Distinctive Talent

So the concept of “credence” proves its worth in rationalising and explaining those actions which are properly seen as governed by decision theory. Do we additionally need a concept of “belief”? Or can credence do everything we need in cognitive psychology?

Perhaps saying “A believes  $p$ ” is simply an imprecise way of saying that A has a high credence in  $p$ ? If this were so, it would explain why “belief” continues to be used in everyday language. But it wouldn’t vindicate it as a scientific term. As MacFarlane writes, “just as we don’t do physics with vague concepts like tall, we won’t do psychology with belief” (MacFarlane 2025, 849).

Similarly unsatisfying are accounts on which belief serves to simplify cognition and thereby make it computationally tractable (such as Staffel (2019), Ross and Schroeder (2014)). If, as Ross and Schroeder (2014) write, belief in a set of propositions is “a defeasible or default disposition to treat them as true in our reasoning” (Ross and Schroeder 2014, 267), we would face the question of what MacFarlane calls “dual control”: Whether the defaults are in charge or are to be departed from would depend on the risks and stakes of the situation, as captured by credences and utilities. But then it seems like credence-

based cognition has to constantly supervise the simplified reasoning. The required computations would not be reduced, but rather increased, if the two are thought to run in parallel in this way (MacFarlane 2025, 851).

What these two have in common is that they inevitably lead to “belief” and “credence” competing for the same theoretical task—and for a task at which “credence” would inevitably be simply better. If credence reductionism is to be resisted, then, belief needs to find a job “in a different line of work altogether” (MacFarlane 2025, 851); one which makes use of belief’s distinctive skill set.

What is this distinctive skill set? MacFarlane identifies two qualifications in particular which “belief” brings to the table and which “credence” cannot match:

First, belief has truth as its correctness condition. If  $p$  turns out not to be true, then belief that  $p$  was wrong. Whereas even arbitrarily high credence in  $p$  may not be proven wrong by  $p$  being shown not to obtain—even credence 1 does not, strictly speaking, rule out  $\neg p$ . As Williamson (2020) points out with the example of a (countably) infinite series of coin tosses: we certainly should have credence 1 in the proposition that at least one of them will come up tails<sup>3</sup>; and this credence will not have shown to have been wrong if—which is possible—not a single toss does come up tails.<sup>4</sup>

Second, belief—unlike credence—can serve as the basis for certain reactive attitudes, as described by Buchak (2014) : I cannot resent you for stealing my phone unless I believe you did. Credence, even an arbitrarily high one, is not enough.

With these talents, belief shouldn’t try to compete with credence in explaining why individuals choose one permitted action over another. They qualify it for a different job: ascribing beliefs to individuals is crucial for the description of another realm of human activity, namely the social practice of reasoning (MacFarlane 2025, 855).

We ascribe to an agent the belief that  $p$  if  $p$  is for them a candidate reason—if, under suitable circumstances, they would assert  $p$  in the language game of giving and asking for reason and thereby assume responsibility for its truth, in the sense of: having reasons for thinking the sentence is true and being willing to provide these reasons upon request.

Believing is crucially intersubjective, and bound up with reasoning about the world in language, in a way in which having a credence is not. I want to illustrate this point from two directions: first, in the way MacFarlane does, who provides broadly cog-sci arguments as to why intersubjective communi-

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<sup>3</sup>Assuming that for each coin toss, we have credence  $1/2$  that it comes up “heads” and  $1/2$  that it comes up “tails”, and that the coin tosses are independent, probabilism commits us to assigning credence  $\frac{1}{2}^n$  to the event “the first  $n$  coin tosses all come up ‘heads’”. As  $n$  approaches infinity, this credence will therefore approach 0 and the credence that at least one of the first  $n$  tosses was “tails” 1.

<sup>4</sup>Or consider selecting at random (sampling from a uniform distribution) a real number in the interval  $[0, 1]$ , and the event “the selected number is rational”. There are rational numbers in this interval; in this sense the event is possible. But the set of rational numbers in  $[0, 1]$  has measure zero; the probability of picking one is therefore 0.

cation requires a coarser grained attitude (compared with an individual's internal credal state). And then, in a more speculative philosophical way, drawing on Donald Davidson, who writes about the connection between participating in a practice of linguistic communication about what is true and the very availability of the concept of objectivity.

First, the MacFarlanian point: the norms of credence, which for him are just the synchronic and diachronic Bayesian requirements of probabilism and conditionalization, “demand an equilibrium of a very large number of mental states”. He theorises that “[p]resumably we can attain this, at least approximately, in our own thought, because we have subpersonal mechanisms that are sensitive to irrationality” (MacFarlane 2025, 858). But in coordinating with others, in settling on a shared picture of the world and a shared sense of what there are and aren't reasons for, “everything has to be under personal-level control” (ibid.). He says to “think of belief and reasons as digital, in contrast to the analog distinctions involved in credence and rationality” (ibid.). The digital version of an analog signal loses some detail, but what it gains is the ability to be communicated efficiently and reliably—“a digital signal can be copied exactly, whereas an analog signal cannot”. In line with this picture, on which the purpose of reasoning is to, in some sense, do a slightly worse version of what an individual's brain does sub-personally, MacFarlane views solitary reasoning as derivative of the more primordial practice of social reasoning: He cites Mercier and Sperber (2017), who write about their experimental results as confirming that people reasoning on their own do so “anticipating a dialogical context, and mostly to find arguments that support their opinion” (Mercier and Sperber 2017, 61).<sup>5</sup>

In a certain sense, MacFarlane does here end up saying that belief simplifies in a suitable way, just like Staffel (2019) or Ross and Schroeder (2014), whose views he rejected earlier; but while e.g. Staffel (2019) points to some complexity-theoretic results like Cooper (1990)'s<sup>6</sup> to argue that the individual's brain cannot be expected to perform the computations required for updating the individual's credences, MacFarlane's argument seems to rather point in the direction of information-theoretic results like the noisy channel theorem, and the fact that, with in a conversation, the number of premises that can be considered and whose reason-relation on a question under consideration can be taken into account is strictly limited in a way that an individual brain's holistic updating procedure need not be.

Donald Davidson (1982), in a much-hated paper which I, however, love, even though it is, objectively, a little bit of a mess, writes on language being required for the having of beliefs. Core to the concept of

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<sup>5</sup>Note that this is compatible with something like the following going on: On a sub-personal level, an individual's brain constantly updates its credence distribution. Whenever an individual is (internally) reasoning in the non-probabilistic way—operating with a small set of premises, weighing whether they support an affirmative judgment on some proposition being considered—they are playing in their head through a potential conversation they might be having with other reasoners. This process may even be influenced through some mechanism (which we are not concerned with here and thus leaving unspecified) by the sub-personal credence distribution represented in the individual's brain.

<sup>6</sup>Which she reports as “probabilistic inference, including conditionalization, is NP-hard”. What Cooper (1990) in fact shows is that general probabilistic inference is NP hard if the probability distribution is represented in the form of a Bayesian network. She does not cite any evidence to the effect that the human brain does in fact represent its credence distribution in this particular way.

belief, he argues, is that it is a state of an entity that can be true or false, correct or incorrect (it differs, I might add, from credence in that way). Essential to the concept of belief is the contrast between what is subjective and what is objective (or, which to Davidson is the same, intersubjective). Any being engaged in linguistic communication has this command of the subjective-intersubjective contrast (and therefore of the concept of truth): for having a conversation with another person entails having a concept of a shared world which both conversation partners are talking about. To understand the other person's utterances, I must think of her world as being the same as my world. Communication requires that my sentences and the sentences of my interlocutor have the same subject matter; and disagreement is only possible against a backdrop of many shared beliefs about that subject matter. Having a concept of such a shared world is the same as having a concept of truth and falsity of the internal states that correspond to one's linguistic utterances, since a belief that  $p$  is true iff  $p$  (iff  $p$  is the case in the world; and, in Wittgenstein's words, if this world were only my world, "whatever is going to seem correct to me is correct. And that only means that here we can't talk about 'correct'" (Wittgenstein 2009, sec. 258) ). To illustrate the situation of a being which does not participate in such an activity Davidson paints the following picture: imagine you are fixed in place and unable to move. It is thus impossible for you to determine the distance of many objects. You know on which line through yourself they are, but not at what distance—it would be impossible to distinguish between certain small objects which are close by and large objects which are far away. Not being fixed in place, you are able to triangulate. From the difference in the way an object appears to you when you move a bit to the left or the right, you can infer how far it is from you. Now, he suggests that our sense of objectivity is itself an effect of an ability to triangulate; not triangulating a spatial object qua spatial object by moving in space, but triangulating an object qua object by accessing another subject's perspective on it by talking about it with them. The very distinction of thing-for-me and thing-for-everyone (i.e. object in the proper sense) is only opened up by interacting with another subject in a particular way: communicating with them about the object in language. (Davidson 1982, 105)

It is this activity, the activity which human beings engage in with each other to construct a shared objective world about which true and false things can be said, that makes the addition of the concept of "belief" to our psychological vocabulary necessary: while the behaviour of human beings as utility maximizers can be described and explained using credence alone, the behaviour of human beings as social reasoners can not. The making of an assertion is an assumption of responsibility for the truth of the asserted content (where this responsibility includes having reasons for thinking the sentence is true and being willing to provide and discuss them); and we use the concept of "belief" to describe and explain what kind of assertions someone is disposed to make.

Based on this account of belief, we can also explain why belief, unlike credence can serve as the basis for reactive attitudes: What certain reactive attitudes (like resentment, blame, pride or shame) require is

precisely a reason. Unlike some other moods (like sadness), for these attitudes it always makes sense to ask (a version of): “Why do you feel proud?” / “Why do you blame Jones?”—and it would be incoherent to reply something along the lines of: “No reason, I’m just feeling proud” (MacFarlane 2025, 856).

## 2.3 The Respective Normativities of Belief and Credence

The above is an outline of when we should ascribe beliefs and credences to an agent; it doesn’t yet say anything about what beliefs or credences an agent should have, or, conversely, under which circumstances an agent can be criticised as unreasonable for the beliefs she holds or as irrational\* for her credences.

In line with our basic methodological behaviourism, we have defined these terms with reference to the kinds of behaviours which are their characteristic expression; action guided by decision theory for credence, and moves in the language game of giving and asking for reasons for belief. For each of these areas of human activity, there is a pre-existing normativity governing them. I will argue that the internal states of belief and credence inherit the normative status of the actions which are their characteristic expression: Since credence just is that internal state which makes one choose certain actions in certain situations, we will call it irrational\* if the actions it makes the agent choose are irrational\*. Likewise for belief: Since to believe  $q$  is to treat  $q$  as a candidate reason in conversation, believing  $q$  is irrational if it is unreasonable to treat  $q$  as a candidate reason.

The paradigm example of an irrational\* action I will consider to be: Entering into a set of bets which the agent recognises to jointly guarantee a loss of utility. It can never be rational\* to give away utility for nothing, if keeping it is also permitted. Consequently, a set of credences is in any case irrational\* if it makes one susceptible to a Dutch Book.

The language game of giving and asking for reasons includes a number of norms; for the application of the framework I am about to discuss, one is particularly relevant: I assume that it includes a norm like: “If you asserted  $p$  yesterday, and had good reasons for it, and have not become aware of any new reasons bearing on the truth of  $p$ ; then you are still committed to  $p$ .”

## 2.4 Application: Sleeping Beauty

In this section, I want to show the power of the above approach by using it to shed light on a philosophical controversy which has been flourishing for decades—due to, as I will argue, a failure to properly distinguish the normativities of belief and credence: What is known as the “Sleeping Beauty” problem.

The paper is first introduced to the philosophical literature by Adam Elga (2000) . It goes as follows:

It is Sunday and Sleeping Beauty (SB) is about to be put to sleep. He<sup>7</sup> will be woken up on Monday,

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<sup>7</sup>In the literature, SB is usually referred to with female pronouns. I will opt to switch that up here. The epistemological



briefly interviewed, and then put back to sleep. A fair coin will be flipped: if it comes up “tails”, a memory loss potion will be administered, which will make SB forget Monday’s events, and he will be woken up again on Tuesday and interviewed again. SB is told all this. If the coin comes up “heads”, SB will not be woken up on Tuesday, but will sleep until Wednesday, when he is woken up in either case, marking the end of the experiment.

The question is: Upon being woken up, what should SB’s stance be on the probability of the coin flip coming up “heads”? And what should it be after he is told, on Monday, that it is, in fact, Monday?

Two paradigmatic positions are staked out by Adam Elga (2000) and David (“Kellogg”) Lewis (2001), respectively:

Lewis’ argument is based on what he dubs his premise (L1), stating that only relevant evidence produces a change in credence; and that SB receives no such evidence regarding “heads” by being woken up. Therefore, his credence in “heads” ought to remain what it was on Sunday:  $1/2$ .

Elga’s argument (in Lewis’ reconstruction) rests on what he calls premise (E1): that SB’s credence in “heads” definitely has to be  $1/2$  after he is told that it is Monday, since in this case, the SB’s credence concerns the flip of a fair coin taking place in the future. Reasoning backwards from this, Elga shows that the evidence of being told that it is Monday can only lead to SB’s credence, after updating on it, being  $1/2$  if it was  $1/3$  prior to receiving this piece of information.

Lewis’ explanation of his disagreement with Elga is as follows: Elga reaches (E1) by applying the principal principle (PP) (Lewis 1981); but, says Lewis, it is here not applicable. PP says that credences should conform to known chances unless one has access to “inadmissible” evidence, such as a reliable prophet’s testimony about the outcome of a future chance event. Lewis agrees that SB, after being told that it is Monday, knows the chance of “heads” to be  $1/2$ ; but, he says, PP is not applicable because being told that it is Monday constitutes evidence from the future for SB—“namely that [s]he is not now in it” (Lewis 2001, 175).

On the picture I have been painting, however, there is a principle more fundamental than the principal principle governing our credences: the principle of avoiding Dutch Books (DB), stemming from the close connection we postulate between credences and decision theory. It is to this principle that the principal principle has to answer, and to which it must defer in case of disagreement—so if DB already narrows down our credences to a specific value, PP need not be consulted at all.

And in fact, this is the case in SB. Hitchcock (2004) presents a Dutch Book against a SB whose credence in “heads” is  $1/2$  (and, in fact, against any SB whose credences differ from  $1/3$ ; but for simplicity’s sake I will here only recount the former). It works as follows<sup>8</sup>:

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upshot of the case depends only on its structural features and will hence be unaffected by this switch.

<sup>8</sup>To make sure that Bookie is not making use of any information advantage she has over SB, we assume that Bookie is undergoing the same treatment as SB—being put to sleep, woken up, receiving the memory loss potion etc.

On Sunday, Bookie offers SB a bet on “tails” whose payout is 30 and whose price is 15. This will be considered fair by SB since his credence in “tails” on Sunday is  $1/2$  (“bet 1”).

Any time SB and Bookie are woken up, Bookie offers SB a bet on “heads” whose price is 10 and payout 20. This will be considered fair by SB since we assume his credence in “heads” upon being woken up is  $1/2$ . On the guaranteed waking up on Monday, this will in any case result in a bet being made (“bet 2”); and if the coin lands “tails”, another bet with the same conditions will be agreed to on Tuesday (“bet 3”).

For the payout structure of this set of bets see Table 2.

Table 2: Hitchcock’s Dutch Book

| Bets                  | Payout      | Cost        | Net gain/loss if... |       |
|-----------------------|-------------|-------------|---------------------|-------|
|                       |             |             | heads               | tails |
| bet 1                 | 30 if tails | 15          | -15                 | 15    |
| bet 2                 | 20 if heads | 10          | 10                  | -10   |
| bet 3 (only if tails) | 20 if heads | 10 if tails | 0                   | -10   |
| Combined              | 20 if heads | 25 if heads | -5                  | -5    |
|                       | 30 if tails | 35 if tails | -5                  | -5    |

Bradley and Leitgeb (2006) argue that Hitchcock’s argument should rather be interpreted as showing that in *Sleeping Beauty*, credences and betting odds come apart<sup>9</sup>. Assessing their argument from the perspective of the above account of belief and credence is illuminating: where for them, it is the epistemic interpretation of credence that must stand firm, to the extent that in cases of conflict, they would rather give up the link between credence and betting, I would maintain that it must be the other way around: Since the job we assigned to credence is specifically to govern our decision-theoretic choices, it should not be interfered with in that realm. But another one of Bradley and Leitgeb’s examples shows how drastic the implications of following through on the strong separation of reasoning and rationality (with truth operating natively as a correctness condition only in the realm of the former but not the latter):

Forgery: A fair coin is about to be flipped. If it lands “heads”, a third party will replace both your and Bookie’s money with fake money, without either of you noticing. If it lands “tails”, you both keep your real money. Bookie will offer to sell you a bet on “heads”, whose payout is 20 of the dollars currently in

<sup>9</sup>Another argument to the same effect is given in Seidenfeld, Schervish, and Kadane (1990), though they do not directly relate it to SB.

her wallet, for the price of 10 of the dollars currently in yours. The outcome of the coin toss is revealed once the deal is struck or refused.

Clearly, you should not accept the bet: the only case in which real money is at stake is the one where you are sure to lose it. Yet, this should not change your belief that the probability of “heads” is  $1/2$  one bit (Bradley and Leitgeb 2006, 123).

Carrying over this reasoning to Sleeping Beauty, they write: “the fact that the agent only avoids a Dutch Book by betting as if her credence in Heads is  $1/3$  gives her no reason to actually believe that the probability of Heads is  $1/3$ ”. I completely agree, SB should not believe that the probability of “heads” is  $1/3$ —but her credence in “heads” has to be  $1/3$  (which Bradley and Leitgeb cannot allow, since they do not distinguish credence and belief in the way we do).

What Bradley and Leitgeb show is that we have to do one of the following two: Either, we have to give up the link between credence and betting; or the link between credence and believing. They do the former; on the account we have been developing, we clearly have to opt for the latter.

What we also have to do is give both Lewis, and Bradley and Leitgeb, their due when it comes the question of what SB should believe. Unlike credence, belief on our account is governed by norms requiring us to keep believing that the probability of “heads” is  $1/2$  both in Forgery and in Sleeping Beauty.

Lewis argued that SB’s credence upon being woken up should be  $1/2$ , because credences are only allowed to change upon receiving relevant evidence—which SB does not. But according to Section 2.3, this is the case only for beliefs. Changing my credences however I see fit does not make me irrational, as long as it doesn’t expose me to a Dutch Book. And in fact, if only a change saves me from a Dutch Book, rationality requires me to make it, no questions asked. Usually, changing my credences in any way which predictably under- or overshoots what would be required by conditionalisation (as SB does when he changes his credence in “heads” to  $1/3$  upon being woken up) would expose me to a Dutch Book (Lewis 1999). But in Sleeping Beauty, it happens to be the other way around: the only way to avoid a Dutch Book is to set my credence to  $1/3$  upon being woken up—so that is what a rational SB must do.<sup>10</sup>

Belief is the state which really isn’t allowed to change without good reason. Since reasoning, and, correspondingly, belief, aim at truth, only emergence of relevant new reasons (which have a bearing on the truth of a belief) should lead to a change in belief. And SB is not made aware of any such reasons between going to sleep on Sunday and being asked the probability of “heads” upon being woken up. Therefore, he should hold steadfastly to his belief that the chance of the coin flip coming up “heads” is

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<sup>10</sup>Why doesn’t the recipe Lewis gives us in Lewis (1999) give us a Dutch Book against a thirder-SB here? Lewis’ diachronic Dutch Book against someone whose credence in  $A$  upon learning  $E$  is predictably different from  $C(A \mid E)$  exposes them to a sure loss of exactly  $C(\neg E)$ —that is, corresponding to the credence they initially assigned to the negation of the evidence they should be conditionalising on. But in Sleeping Beauty, the evidence on which SB is “updating incorrectly” is “I am woken up” (or “It is either Monday and ‘tails’, or Monday and ‘heads’, or Tuesday and ‘tails’”). But their credence in not being woken up / in it being none of these three, is zero. So Lewis’ Dutch Book exposes the thirder-SB to a sure loss of exactly \$0.

1/2.

So: If the question SB is asked is: “What do you believe the probability of ‘heads’ to be?”, his answer should be “1/2” (he should of course also stick to this assessment after being told that it is Monday; that is, when the question is framed as being about SB’s beliefs, the “double halfer” position is correct; cf. Titelbaum (2013), Meacham (2008)). If, however, SB is asked about his credence, or offered to bet, then he needs to quote 1/3.

## 2.5 In Search of A Pineal Gland

MacFarlane, in summarising the division of labour he has in mind between the concepts of “belief” and “credence”, emphasises the relative independence between rationality and reasoning:

Reasoning is the process of producing, assessing, and criticizing reasons. So understood, reasoning has only a tenuous relation to rationality. Rationality, as I understand it, is a matter of the internal coherence of one’s attitudes, both synchronically and diachronically. This is what Bayesian epistemology and decision theory attempt to give an account of. One can be rational without ever reasoning, and without ever regarding anything as a reason. Conversely, reasoning—even good reasoning—does not necessarily make one rational. (MacFarlane 2025, 855)

Putting it this way, we are of course presented with the question—what is reasoning good for? MacFarlane’s answer: “Coordinating with others” (MacFarlane 2025, 863) which, as discussed above, requires suitable coarse-graining.

But part of the story is missing: How does the coordinating affect the individual’s actions? MacFarlane seems to be in a precarious position here: to the extent that the result of the communication is the formation, in the individual, of new beliefs, these beliefs should never feature in an explanation of that individual’s actions—that is credence’s job! And belief should not interfere with it, since it’s talents do not qualify it for this job. But a story of how beliefs affect credences is lacking—and in fact the above quote seems to suggest that, in general, they do not.

Well, somehow they have to. There needs to be a pineal gland which connects the realm of credence and the realm of belief, on pain of casting the whole practice of reasoning as an epiphenomenon.

In this section, I want to suggest such a “pineal gland”, a point where belief and credence do meet; I want to suggest that they do in the practice of betting. The result will be a view where reasoning is in a sense primary: certain choices of an agent will be seen as a reason to call that agent irrational, and irrational is something one should not be; thus reducing the normativity governing credences to the power of belief-based reasoning to determine what one should and shouldn’t be or do.

A natural point to start when looking for a point where beliefs affect credences is conditionalisation, which may be framed as saying: If you have the belief that  $p$ , set your credence in  $q$  to  $Cr(q \mid p)$ . A drawback of just running with this is that it would seem to entail that one ought to have credence 1 or 0 in anything which one has a belief about (since probabilism requires  $Cr(p \mid p) = 1$ )—a view which is widely rejected, and, I think, for good reasons. It seems perfectly fine to assert something without thereby committing oneself to betting on it at any odds.

[...some kind of transition...]

Successful execution of a bet between two people from start to finish involves and connects both decision theory and reasoning, both credence and belief (I take general latent awareness of this fact to be the reason why Dutch Book arguments are often used to convince people of Bayesianism, and, as I will expand on later, are in fact the only correct justification of it). An individual's action within such a process is guided, at times, both by decision theory and by the normative requirements to which one is subject as a reasoner (and thus, believer). The way these are connected through a (real or imagined) bet will impose indirect epistemic requirements on credence, even if the only direct normative requirements on credence are pragmatic.

Here, I understand “betting” as the social practice where two parties belonging to the same linguistic community enter into an agreement with the content that an exchange of a set amount money should take place—usually at a future point in time which is at least loosely specified—conditional on the truth of a certain statement. This can (but doesn't have to) take the form of one party selling the other a “bet” in the sense defined in the introduction.

For a specific  $q$ , any such agreement is a bet on  $q$  iff the parties share an understanding that the pot should be awarded to the person who bet on  $q$  iff  $q$ . By entering into a bet on  $q$ , I commit myself to the this proposition about who should receive the pot under which conditions. When it is time to settle the bet, in order to settle it, the parties must reach agreement on whether  $q$ ; that is, they must reach a state where both hold the same belief about  $q$ .

Based on this understanding of a bet, I want to propose what I call the Betting Norm, which I take to be more fundamental than the “Dutch Book” norm (and, in fact, explaining it). It concerns the situation where an agent is offered a bet, or a set of bets, and it specifies one way in which accepting such a (set of) bet(s) can be irrational\* for her; namely, if she already expects that any reasonable person will in the end reach the judgment that the bets are to be settled in a way which would entail that she overall loses money.

Betting Norm: Assume you are invited to enter into a (set of) bet(s), and the outcome of this interaction affects your utility only via the amount of money you gain or lose, and is strictly monotonically increasing in the amount of money you gain. Then: It is irrational\* for you to

accept, if you already believe for some  $\phi$  that  $\phi$  will be the only reasonable thing to believe at the time when the bet(s) are to be settled, and  $\phi$  being true implies that the total payout of the (set of) bet(s) should be such that you lose money.

[...spell out how Dutch Books are of the form “if  $\top$ , you lose money”, so that even though you do not know which contingent proposition will be the only reasonable one to believe, you do know that it will be untenable to deny  $\top$ , so that the Betting Norm explains why you should not enter into Dutch books...]

[... spell out how this motivates conditionalisation without committing one to the above view that credences in believed propositions have to be 1; this will work via the distinction between “I believe that  $p$ ” and “I believe that  $p$  will be the only reasonable thing to believe” (i hope)...]

### 3 Basing Reasons and Attitudes on Prediction Market Prices

[...under construction...]

## 4 Prediction Markets

### 4.1 Betting Against The Market

[...under construction]

### 4.2 (Rough) Argument(s) for: Belief Should Not Usually be Affected by Market Prices of Bets

[...under construction...]

## 5 A Note on the Practical Implications of this View for Individual and Policy Decision Making

What does this mean for prediction markets as a potential source of information in an individual's, or a government's decision making? It is a consequence of this view that market prices of bets cannot legitimately be used for decisions which are matters of justice, or where anything other than simple maximisation of utility is at stake. But insofar as maximisation of utility is the overriding dimension along which a decision is to be evaluated, prediction markets are suitable for eliciting probabilities to be used in expected utility calculations.

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